

Trading days effects

Definition

For each period t , the days are divided in K groups $\{D_{t1}, \dots, D_{tK}\}$.

The groups of days can be anything (trading days, working days, Sundays + holidays assimilated to Sundays...)

We write $N_t = \sum_{i=1}^K D_{ti}$, the number of days of the period t

The effect of one day of the group i is measured by α_i , so that the global effect of the group i for the period t is $\alpha_i D_{ti}$

The global effect of all the days for the period t is

$$\sum_{i=1}^K \alpha_i D_{ti} = \bar{\alpha} N_t + \sum_{i=1}^K (\alpha_i - \bar{\alpha}) D_{ti}$$

where $\bar{\alpha} = \sum_{i=1}^K w_i \alpha_i$ with $\sum_{i=1}^K w_i = 1$

So,

$$\sum_{i=1}^K (\alpha_i - \bar{\alpha}) w_i = \sum_{i=1}^K \alpha_i w_i - \bar{\alpha} \sum_{i=1}^K w_i = 0$$

We focus now on $\sum_{i=1}^K (\alpha_i - \bar{\alpha}) D_{ti}$, the actual trading days effects (excluding the length of period effect).

Writing $\alpha_i - \bar{\alpha} = \beta_i$ and using that $\sum_{i=1}^K \beta_i w_i = 0$, we have that

$$\sum_{i=1}^K \beta_i D_{ti} = \sum_{i=1}^K \beta_i (D_{ti} - \frac{w_i}{w_K} D_{tK}) = \sum_{i=1}^{K-1} \beta_i (D_{ti} - \frac{w_i}{w_K} D_{tK})$$

Note that the relationship is valid for any set of weights w_i . It is also clear that the contrasting group of days can be any group:

$$\sum_{i=1}^{K-1} \beta_i (D_{ti} - \frac{w_i}{w_K} D_{tK}) = \sum_{i=1}^{K, i \neq J} \beta_i (D_{ti} - \frac{w_i}{w_J} D_{tJ})$$

The "missing" coefficient is easily derived from the others:

$$\beta_K = -\frac{1}{w_K} \sum_{i=1}^{K-1} \beta_i w_i$$

Correction for deterministic seasonality

In the case of seasonal adjustment, we further impose that the regression variables don't contain deterministic seasonality. That is achieved by removing from each type of period (month, quarter...) its long term average. We write $\bar{D}_i^y$ the long term average of the yearly number of days in the group i and $\bar{D}_{i,J}^y$ the long term average of the number of days in the group i for the periods J (for instance, average number of Mondays in January...).

The corrected contrast for the time t belonging to the period J is:

$$C_{ti} = D_{ti} - \bar{D}_{i,J}^y - \frac{w_i}{w_K} (D_{tK} - \bar{D}_{K,J}^y)$$

Weights of the groups of days

We can define different sets of weights. The usual one consists in giving the same weight to each type of days. w_i is just proportional to the number of days in the group i . In the case of "week days", $w_0 = \frac{5}{7}$ (weeks) and $w_1 = \frac{2}{7}$ (week-ends). In the case of "trading days", $w_i = \frac{1}{7}$... Another approach consists in using the long term yearly averages, taking into account the actual holidays. We get now that $w_i = \frac{\bar{D}_i^y}{365.25}$.

After the removal of the deterministic seasonality, the variables computed using the two sets of weights considered above are very similar. In the case of the "trading days", the difference for the time t , belonging to the period J , and for the day i with contrast K is $(1 - \frac{w_i}{w_K})(D_{tK} - \bar{D}_{K,J}^y)$, which is usually small. By default, JD+ uses the first approach, which is simpler. The second approach is implemented in the algorithmic modules, but not available through the graphical interface.

Use in RegArima models

In the context of RegArima modeling, we can also observe that the global effect of the trading days doesn't depend neither on the used weights (we project on the same space) nor on the contrasting group (see above) nor on the long term corrections (removed by differencing).

The estimated coefficients slightly change if we use different weights (not if we use a different contrasting group). It must also be noted that the choices affect the T-Stat of the different coefficients (not the joint F-Test), which can lead to other solutions when those T-Stats are used for selecting the regression variables (Tramo). Considering that the leap year/length of period variable is nearly independent of the other variables, the test on that variable is not very sensitive to the various specifications.

Conclusion

The use of different specifications of the trading days doesn't impact the final results (except through some automatic selection procedure). It just (slightly) changes the way we interpret the estimated coefficients.

